

Color Animated Full-parallax High-definition Computer-generated Hologram

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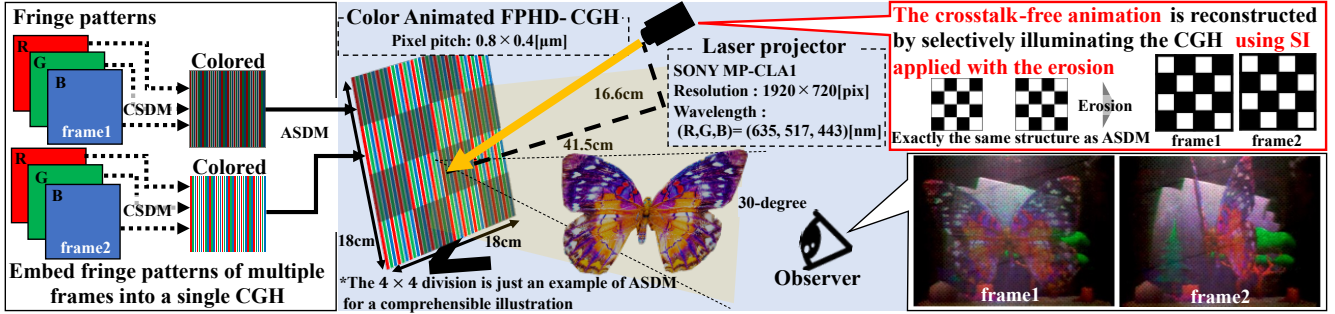


Figure 1: System overview of our two-frame color animation prototype

ABSTRACT

We realized color animation of a computer-generated hologram (CGH) while retaining the features of the conventional full-parallax high-definition CGH (FPHD-CGH) method such as large screen size and a wide viewing area. Since the FPHD-CGH is fabricated using a laser direct writing system, the only way to animate it is to switch structured illumination (SI), and even that method has the problem of overlapping images from different frames caused by light leakage. We propose a method of suppressing this light leakage while maintaining image quality by applying erosion to SI. We also developed the world's first two-frame color animation prototype, which is 18 cm square and has a 30-degree viewing angle horizontally and vertically. We did this by combining our method with the existing FPHD-CGH colorization method.

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1 INTRODUCTION

Computer holography is regarded as the ultimate 3D display technology because it can reproduce physical light emitted from an object by optical diffraction of a digital fringe pattern, a *computer-generated hologram (CGH)*. To make computer holography practical,

both a large screen size and a wide viewing angle are essential, and these conditions require a CGH with an extremely large number of pixels and fine pixel pitch. Achieving this type of CGH display is the greatest challenge in CGH research [Jungkwuen An and Lee 2022; Sahin et al. 2020]. In addition, color animation of CGH is important because it will expand the range of applications, such as digital signage. However, no method currently exists that satisfies all of these requirements.

One of the most promising avenues for realizing a large screen size and a wide viewing angle among the above-mentioned requirements is the full-parallax high-definition CGH (FPHD-CGH) [Matsushima 2020]. Since the FPHD-CGH is fabricated by printing a fringe pattern using a laser direct writing system, it is a static CGH. Thus, animation using FPHD-CGH cannot be realized by switching fringe patterns and does not exist to our knowledge, except for the work of [Matsushima et al. 2021]. In their approach, a single FPHD-CGH is first fabricated by embedding fringe patterns for multiple frames using space division multiplexing (SDM). Then, the fringe pattern areas corresponding to each frame are selectively illuminated using structured illumination (SI) which is pattern illumination turned on or off spatially in exactly the same pattern as SDM. However, their research remained incomplete due to the crosstalk, which is the overlap of reconstructed images caused by light leakage into other fringe patterns, and only monochrome animation was achieved.

To enable perfect animation, we propose a method to suppress light leakage while maintaining image quality by applying erosion to SI. We also developed the world's first color animation prototype while retaining a large screen and a wide viewing angle features of the FPHD-CGH method by combining our method with the existing FPHD-CGH colorization method [Tsuchiyama and Matsushima 2017]. Our prototype achieved a two frame-color animation having a large size (18 cm square) and wide viewing angle (30 degrees

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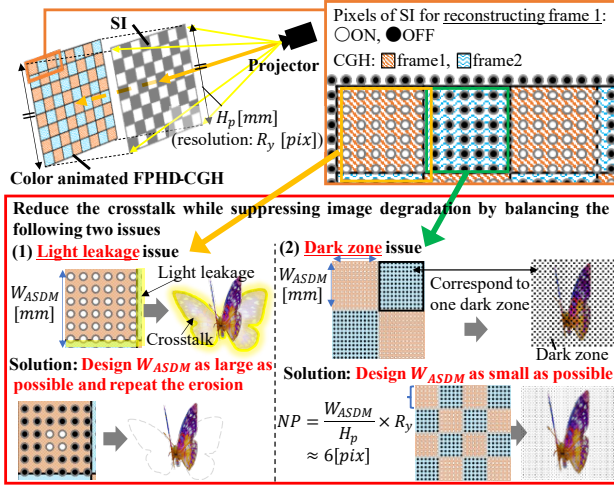


Figure 2: Our crosstalk elimination method

horizontally and vertically), while the viewing angle of the existing maximum size (25.4 cm diagonal) full parallax CGH method is only 15 degrees, and the size of the existing maximum viewing angle (36 degrees) full parallax CGH method is only 6 mm diagonal [Jungkwen An and Lee 2022].

2 PROPOSAL AND IMPLEMENTATION

We propose a method to eliminate the crosstalk while avoiding negative effects on image quality. Light leakage, which causes the crosstalk, inevitably occurs due to the spread of light and slight misalignment of the optical system in practice. We considered that light leakage could be almost completely eliminated by increasing the block width (W_{ASDM}) of the animation SDM (ASDM) and repeating the 1-pixel erosion operation to SI as shown in Fig. 2 (1). However, a large W_{ASDM} makes dark zones, which are non-illuminated CGH areas, more noticeable as shown in Fig. 2 (2). In addition, repeating the erosion causes a decrease in brightness. Thus, we experimentally determined an ASDM design that balances this quality trade-off. We describe the details of our prototype implementation in the following sections, focusing on the follow-up to the aforementioned trade-offs.

2.1 Design of SDM

As shown in Fig. 1, to achieve the color reproduction of the prototype, three fringe patterns corresponding to the RGB color components were integrated by color SDM (CSDM) before applying ASDM. Specifically, CSDM enables color reproduction with a single CGH by attaching RGB color filters with the same CSDM pattern to the fringe pattern made of chromium thin films according to [Tsuchiyama and Matsushima 2017]. We used an 80 μm width strip-shaped CSDM, identical to [Tsuchiyama and Matsushima 2017] because CSDM shapes do not contribute to the aforementioned quality trade-off.

For animation, we adopted a checkerboard-shaped ASDM with the same width horizontally and vertically. This is because our preliminary experiment with different ASDM shapes showed that SI switching with a checkerboard-shaped ASDM was less perceptible than that with a stripe-shaped ASDM. We experimentally

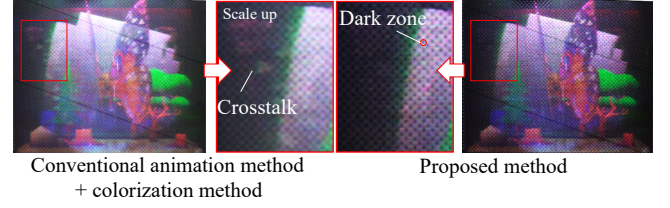


Figure 3: Optical reconstruction of our prototype

determined $W_{ASDM} = 1.68$ mm based on the width of CSDM and the minimum length provided control of light leakage works sufficiently with the projector used in our experiment. The number of pixels (NP) in one block of ASDM is almost 6×6 pixels, based on the projector's resolution (R_y) and the projection size (H_p) as shown in the equation in Fig. 2.

2.2 Implementation of SI

We generated SI using a projector that projects a checkerboard-shaped image consisting of black (illumination OFF) and white (illumination ON) blocks with the same structure as ASDM. To minimize light leakage, we used a scanning laser projector (SONY MP-CLA1). Since illumination light spreads lightly even with the projector, we adjusted the number of erosion repetitions while visually checking the optical reconstruction so as to balance light leakage and dark zones.

3 DISCUSSION AND CONCLUSION

We developed a prototype of the two-frame color animated FPHD-CGH with an 18 cm square display and a 30-degree viewing angle horizontally and vertically as shown in Fig. 1. In addition, the optical reconstruction shown in Fig. 3 indicates that our prototype is capable of animation without the crosstalk while minimizing the occurrence of dark zones. Although we observed slight dark zones in the optical reconstructions, we consider these dark zones would be less noticeable if a higher-resolution projector were to be used. The reason is that a higher-resolution projector can make W_{ASDM} smaller with the same NP.

Our method has the capacity to increase the number of frames, however, simply increasing frames leads to a noticeable dark zone because it makes the illuminated area per frame smaller. We think this issue can be solved by designing SI to irradiate the common structure among frames so that one pattern is not limited to only one frame. As future work, we will study how to find the optimal pattern by considering the correlation among frames.

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